

1022. Sum of Root To Leaf Binary Numbers

Difficulty: Easy

Link: <https://leetcode.com/problems/sum-of-root-to-leaf-binary-numbers/description/?envType=daily-question&envId=2026-02-24>

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Problem:

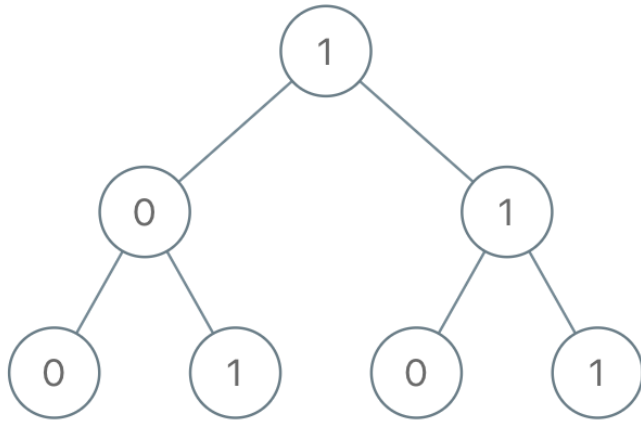
You are given the `root` of a binary tree where each node has a value `0` or `1`. Each root-to-leaf path represents a binary number starting with the most significant bit.

- For example, if the path is `0 -> 1 -> 1 -> 0 -> 1`, then this could represent `01101` in binary, which is `13`.

For all leaves in the tree, consider the numbers represented by the path from the root to that leaf. Return *the sum of these numbers*.

The test cases are generated so that the answer fits in a **32-bits** integer.

Example 1:



Input: root = [1,0,1,0,1,0,1]

Output: 22

Explanation: $(100) + (101) + (110) + (111) = 4 + 5 + 6 + 7 = 22$

Example 2:

Input: root = [0]

Output: 0

Constraints:

- The number of nodes in the tree is in the range [1, 1000] .
- Node.val is 0 or 1 .